

PRE-ENGINEERING PATHWAY

Courses

Below you will find a list of our current or recent offerings. See the course catalog for descriptions and updated information.

CHEMISTRY

COMPUTER SCIENCE

MATHEMATICS

PHYSICS

FUNDAMENTALS OF PHYSICS I & II (53-154, 164) (3-3). A calculus-based treatment of mechanics, wave motion, electromagnetism, optics. Prerequisites: Concurrent registration or credit in Mathematics 52-154. (Physics 53-154 is prerequisite for Physics 53-164.) (NSL)

ELECTRONICS (53-204) (3-3). Introduction to digital and analog circuits, with applications to modern instrumentation. Prerequisites: Mathematics 52-154, 254 and Physics 53-154, 164. Mathematics 52-353 is a pre- or co-requisite for this course. (NSL)

ELEMENTARY MODERN PHYSICS (53-214) (3-3). An introduction to the physics of the 20th century that surveys relativity theory, wave-particle duality, atomic structure, wave mechanics, nuclear theory and particle physics. Prerequisites: Mathematics 52-154, 254 and Physics 53-154, 164. Mathematics 53-353 is a pre- or co-requisite for this course. (NSL)

CLASSICAL MECHANICS I (53-334) (3-3). An advanced treatment of Newtonian mechanics with applications to kinematics, forced oscillations, central force motion and systems of particles. Prerequisites: Mathematics 52-353 and Physics 53-154, 164. Mathematics 53-753 is a pre- or co-requisite for this course. (NSL)